Figures: Hospital Readmission Prediction for Diabetic Patients

COMP0189 Applied AI — Coursework 1 | Candidate: TGSZ1

TASK 1: DATASET DESCRIPTION

1.1 Characteristics

Table 1: Summary statistics for the 50 dataset columns. Numeric features report median and IQR; categorical features report cardinality and categories (up to 10 shown). † Integer-encoded categorical ID; median not meaningful.

Feature	dtype	Median (IQR)	Cardinality	Categories/ Notes
Identifiers (excluded from modelling)				
`encounter_id`	int64	—	101,766	Unique encounter identifier
`patient_nbr`	int64	—	76,099	Unique patient identifier; used as grouping key
Administrative ID features (3) — integer-encoded categorical IDs†				
`admission_type_id`	int64	—	8	range=[1,8]
`discharge_disposition_id`	int64	—	28	range=[1,28]
`admission_source_id`	int64	—	25	range=[1,25]
Numeric features (8)				
`time_in_hospital`	int64	4.0 (4.0)	—	range=[1,14]
`num_lab_procedures`	int64	44.0 (26.0)	—	range=[1,132]
`num_procedures`	int64	1.0 (2.0)	—	range=[0,6]
`num_medications`	int64	15.0 (10.0)	—	range=[1,81]
`number_outpatient`	int64	0.0 (0.0)	—	range=[0,42]
`number_emergency`	int64	0.0 (0.0)	—	range=[0,76]
`number_inpatient`	int64	0.0 (1.0)	—	range=[0,21]
`number_diagnoses`	int64	8.0 (3.0)	—	range=[1,16]
Categorical features — individual (13)				
`race`	object	—	6	Caucasian, AfricanAmerican, ?, Hispanic, Other, Asian
`gender`	object	—	3	Female, Male, Unknown/Invalid
`age`	object	—	10	[0-10), [10-20), [20-30), [30-40), [40-50), [50-60), [60-70), [70-80), [80-90), [90-100)
`weight`	object	—	10	?, [0-25), [25-50), [50-75), [75-100), [100-125), [125-150), [150-175), [175-200), >200
`payer_code`	object	—	18	18 unique values
`medical_specialty`	object	—	73	73 unique values
`max_glu_serum`	object	—	3	None, Norm, >200, >300
`A1Cresult`	object	—	4	None, >7, >8, Norm
`change`	object	—	2	No, Ch
`diabetesMed`	object	—	2	Yes, No
`diag_1`	object	—	717	717 ICD-9 codes
`diag_2`	object	—	749	749 ICD-9 codes
`diag_3`	object	—	790	790 ICD-9 codes
Medication dosage indicators (23), grouped				
Medication dosage indicators (23)	object	—	1–4	No, Steady, Up, Down (subset per drug)
Target				
`readmitted`	object	—	3	NO, <30, >30

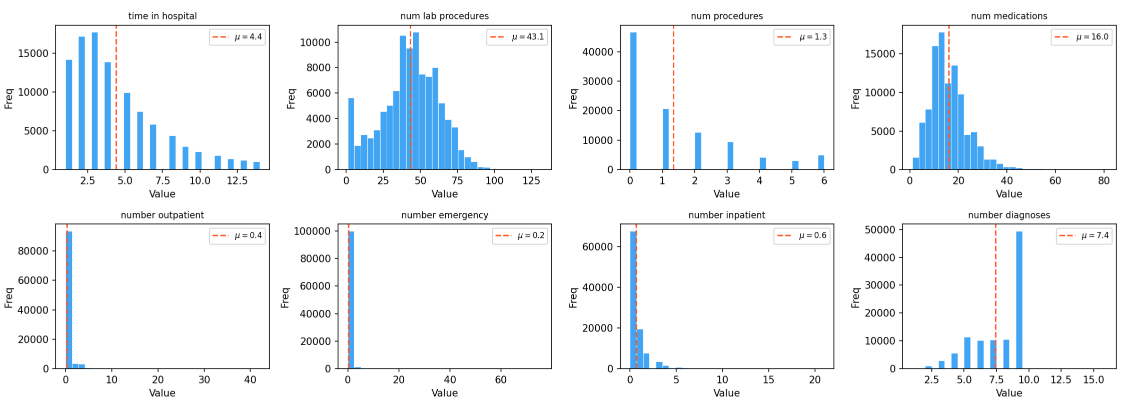

Figure 1: Distribution of the eight numeric features in the dataset (integer-encoded categorical ID features are excluded here).

Table 2: Raw missingness indicators for categorical features. Numeric features contain no missing values.

Categorical Feature	Missingness Indicator
A1Cresult	"None"
max_glu_serum	"None"
admission_type_id	[5, 6, 8] (Not Available, NULL, Not Mapped)
discharge_disposition_id	[18, 25, 26] (NULL, Not Mapped, Unknown/Invalid)
admission_source_id	[9, 15, 17, 20, 21] (Not Available, Not Available, NULL, Not Mapped, Unknown/Invalid)
gender	"Unknown/Invalid"
medical_specialty	"?"
payer_code	"?"
race	"?"
weight	"?"
diag_1	"?"
diag_2	"?"
diag_3	"?"

Figure 2: Distribution of the main categorical features in the dataset. High cardinality features that are encoded or dropped (diag_1, diag_2, diag_3; medical_specialty; payer_code) are not excluded for clarity.

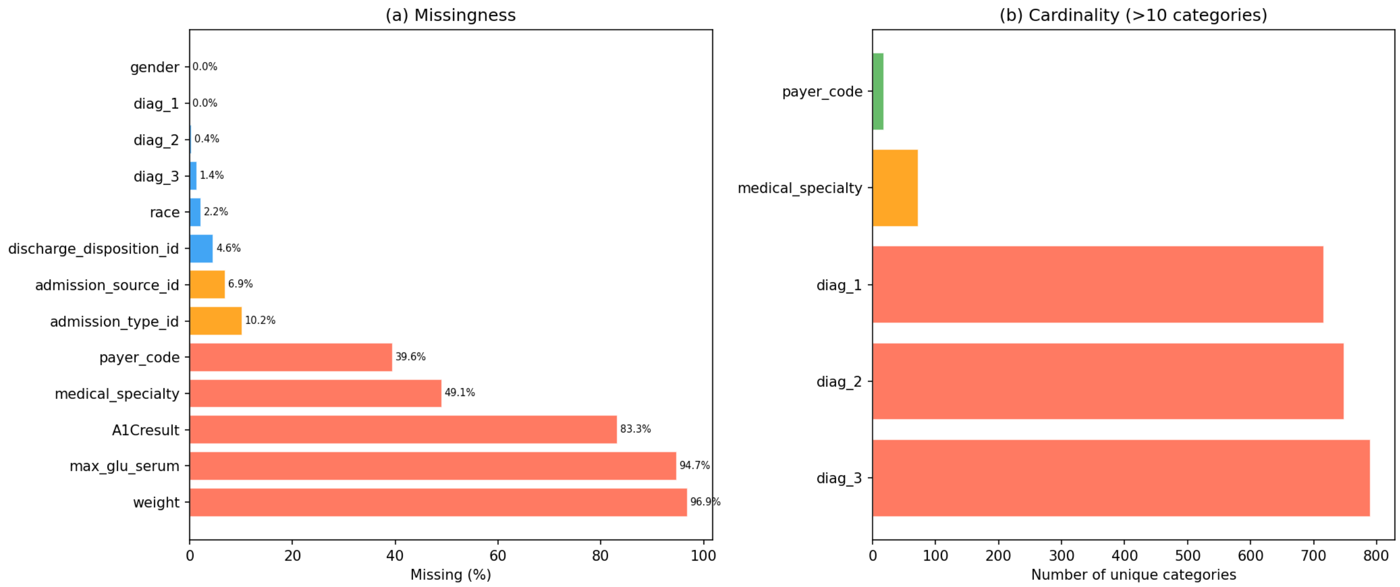

Figure 3: Percentage missingness by feature and number of unique categories for features with more than 10 categories.

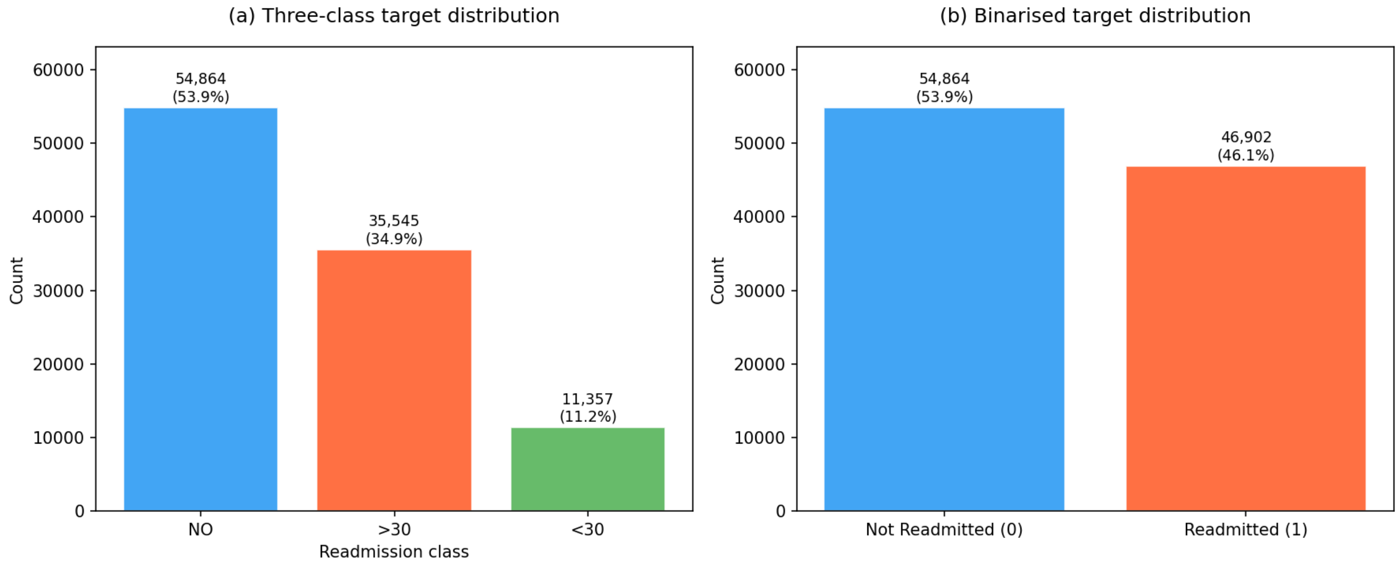

Figure 4: Class imbalance of target distribution before and after binarisation.

1.2 Challenges

Table 3: Missingness summary and treatment strategy. [†]Integer-encoded categorical ID; sentinel codes recoded to `NaN` prior to pipeline. [‡]Structurally absent when `number_diagnoses` $< 2/3$; not randomly missing. [§]Clinically informative; active decision not to test. [¶]Binary encoding `(gender == 'Male').astype(int)` implicitly imputes as Female (mode); all 3 records are absent from the modeling dataset. Mode imputation used throughout as all imputed features are categorical.

Feature	Miss%	Mechanism	Imputation	Mask?
Dropped				
`weight`	96.9%	MNAR	Drop	—
`medical_specialty`	49.1%	MAR	Drop	—
`payer_code`	39.6%	MAR	Drop	—
Recoded — not treated as missing				
`max_glu_serum`	94.7%	N/A[§]	`None` $\rightarrow$ `Not_Tested`	No
`A1Cresult`	83.3%	N/A[§]	`None` $\rightarrow$ `Not_Tested`	No
`diag_2`	0.4%	Structural[‡]	`None` category	No
`diag_3`	1.4%	Structural[‡]	`None` category	No
Mode imputed within CV folds (all categorical)				
`admission_type_id`[†]	10.2%	MAR	Mode	Yes
`admission_source_id`[†]	6.9%	MAR	Mode	Yes
`discharge_disposition_id`[†]	4.6%	MAR	Mode	Yes
`race`	2.2%	MNAR	Mode	Yes
`diag_1`	0.02%	Negligible	Mode $\rightarrow$ ICD-9 mapping	No
Implicit mode imputation via encoding				
`gender`[¶]	0.003%	Negligible	Mode (implicit)	No

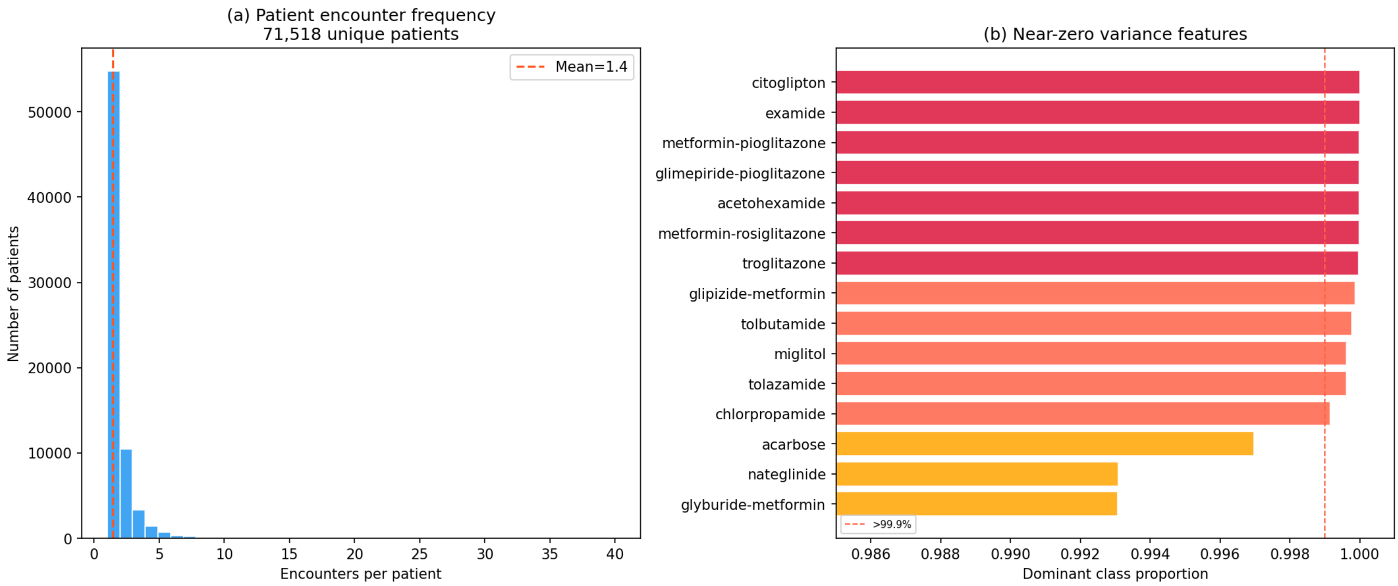

Figure 5: Plots of patient encounter frequency and feature variance for features with near-zero variance (>99.9% one class)

TASK 2: DATA ASSEMBLING & PRE-PROCESSING

2.1 Assembly

Table 4: Values of `discharge_disposition_id` corresponding to patient death or hospice discharge.

`discharge_disposition_id`	**Description**
11	Expired
19	Expired at home. Medicaid only, hospice
20	Expired in a medical facility. Medicaid only, hospice.
21	Expired, place unknown. Medicaid only, hospice.
13	Hospice / home
14	Hospice / medical facility

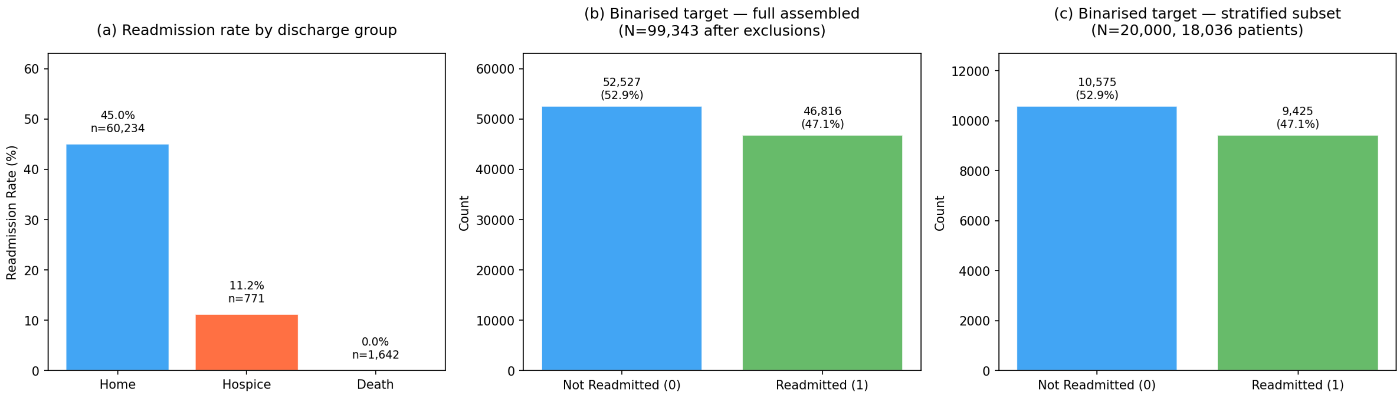

Figure 6: Plots of readmission by discharge group, of binarised target distribution after exclusion of hospice and death discharges, and of binarised target after stratified subsampling.

2.3 Diagnosis Encoding

Table 5: ICD-9 clinical category mapping applied to `diag_1/2/3`. Based on Strack et al. (2014). †Appears in `diag_2/3` only.

Category	ICD-9 Code Range
Diabetes	250–250.99
Circulatory	390–459, 785
Respiratory	460–519, 786
Digestive	520–579, 787
Injury	800–999
Musculoskeletal	710–739
Genitourinary	580–629, 788
Neoplasms	140–239
Endocrine/Metabolic	240–249, 251–279
Blood Disorders	280–289
Infectious/Parasitic	1–139
E/V Codes	E-codes, V-codes
Other	Unclassified or unknown
No Diagnosis†	Structural absence (n=358, n=1,423)

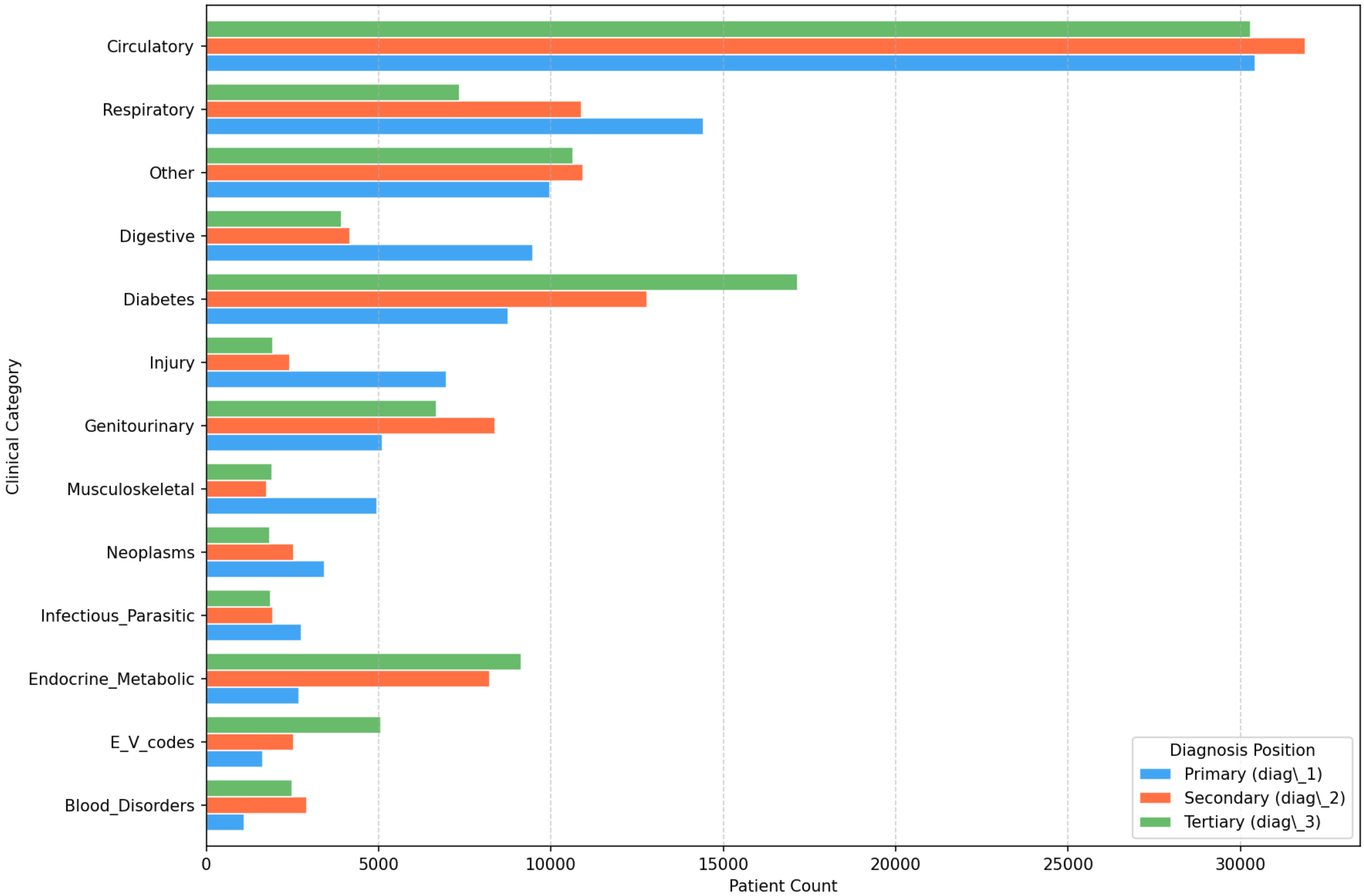

Figure 7: Distribution of ICD-9 clinical categories across primary, secondary, and tertiary diagnosis fields after mapping to 13 categories (§2.3, Table 5). `No_Diagnosis` (n=358 for `diag_2`; n=1,423 for `diag_3`) is excluded as it reflects structural absence rather than a clinical category. `Other` includes unclassified ICD-9 codes; `diag_1`'s 21 missing values (0.02%) are mode-imputed prior to mapping and do not appear here.

3.4 Hyperparameter Analysis

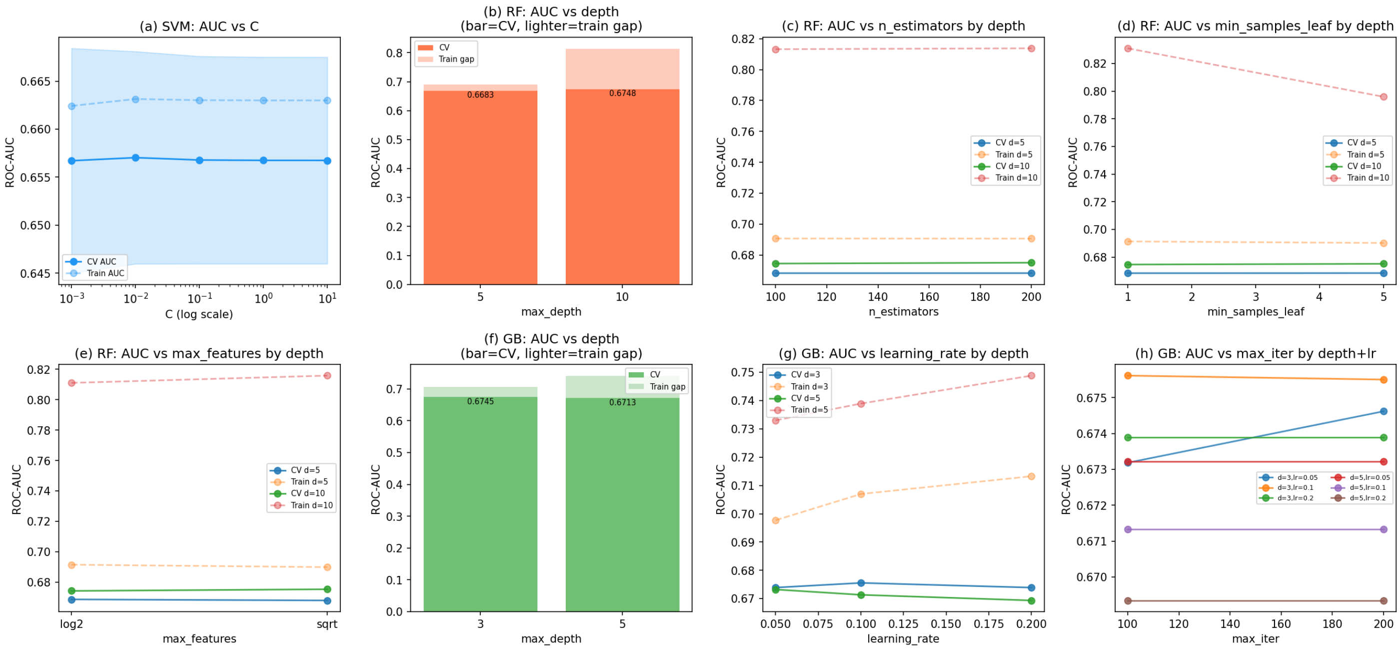

Figure 8: Plots showing CV and train ROC-AUC across all hyperparameter combinations from a full 20,000-sample `GridSearchCV` refit. Solid lines and bars show CV ROC-AUC; dashed lines and lighter bar segments show train ROC-AUC, with the gap indicating overfitting degree.

3.6 Discussion of Results

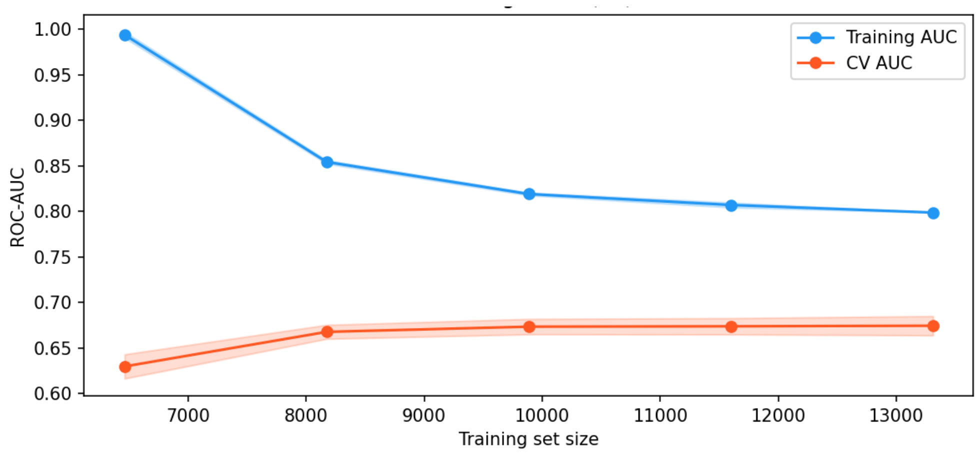

Figure 9: Learning curve for the best RF estimator, evaluated using `StratifiedGroupKFold` ($k=3$, `groups=patient_nbr`) across eight evenly-spaced training set sizes from 10% to 100% of the 20,000-sample subset. Shaded regions show ± 1 SD across folds.

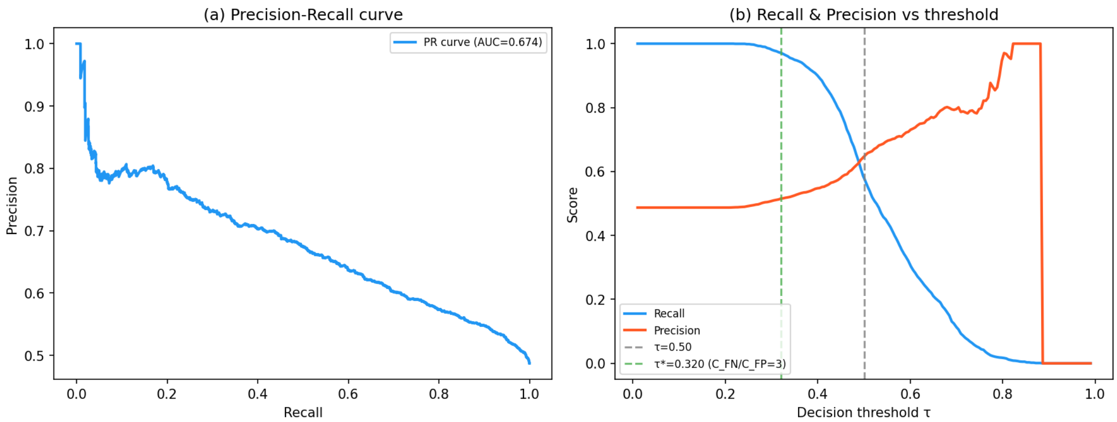

Figure 10: Threshold analysis for the best RF estimator evaluated on outer fold 1. **(a)** Precision-Recall curve with PR-AUC. **(b)** Recall and Precision as a function of decision threshold τ, with the default threshold $\tau=0.50$ and the cost-sensitive optimal threshold τ^* (derived by maximising $3\times$TPR $-$ FPR, corresponding to $C_{\text{FN}}/C_{\text{FP}}=3$) shown as vertical dashed lines.

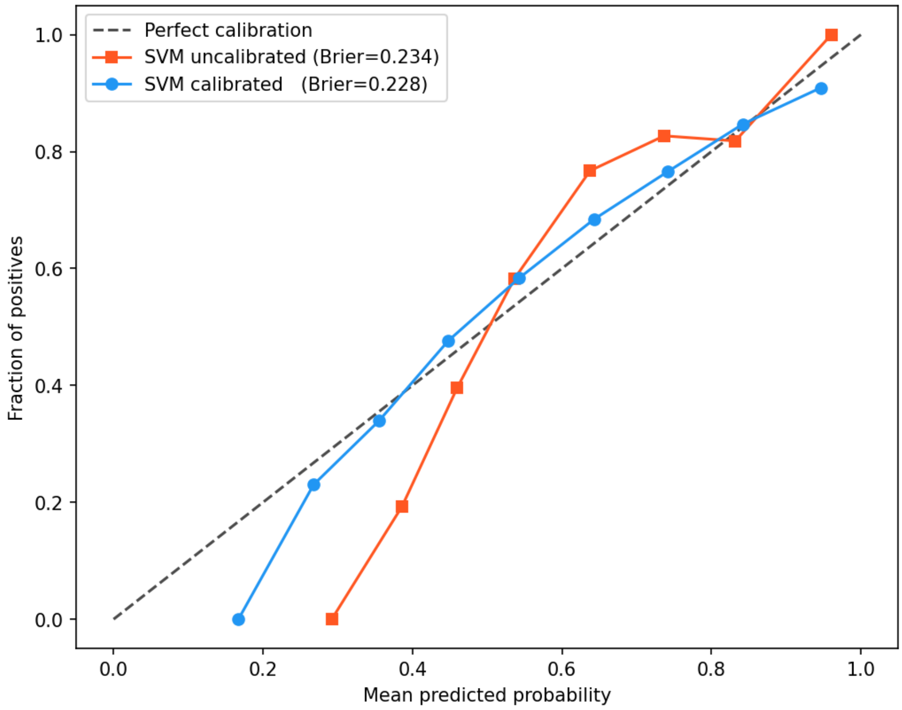

Figure 11: Calibration curves for the SVM before and after `CalibratedClassifierCV` (sigmoid, $k=3$), evaluated on outer fold 1. Uncalibrated probabilities are derived by applying the sigmoid function to raw `LinearSVC` decision function scores. Brier scores are shown in the legend; the dashed diagonal represents perfect calibration.

TASK 4: MODEL INTERPRETATION

4.1 Feature Importance

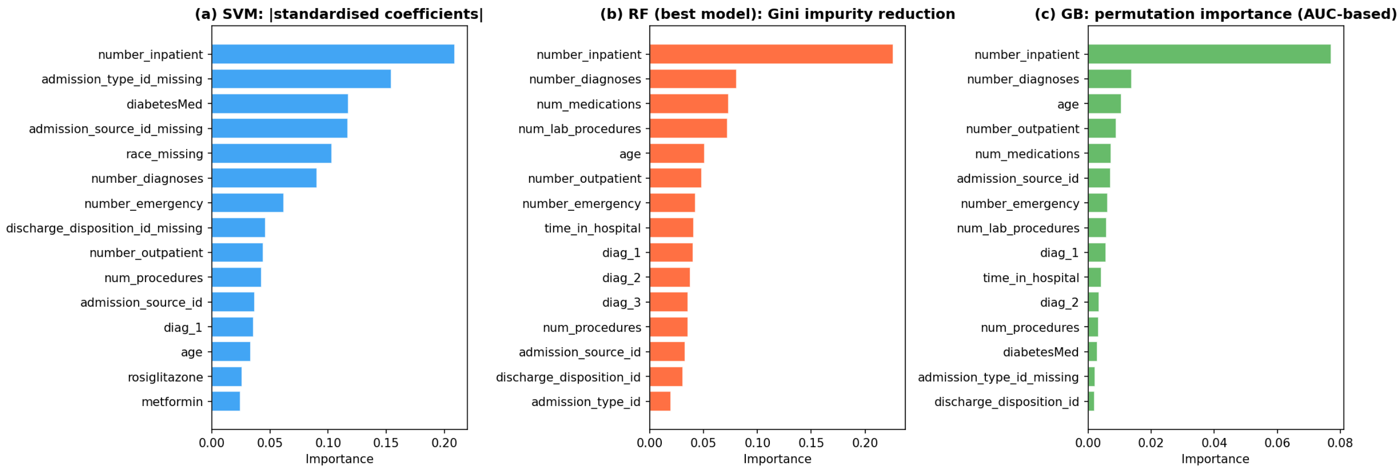

Figure 12: Native feature importance for the top 15 features per model, computed on the 20,000-sample stratified subset. **(a)** SVM: absolute standardised coefficients from the `LinearSVC` estimator. **(b)** RF: mean decrease in Gini impurity. **(c)** GB: permutation importance (ROC-AUC based, 10 repeats); `HistGradientBoostingClassifier` does not expose impurity-based importance. Negative permutation values are clipped to zero.

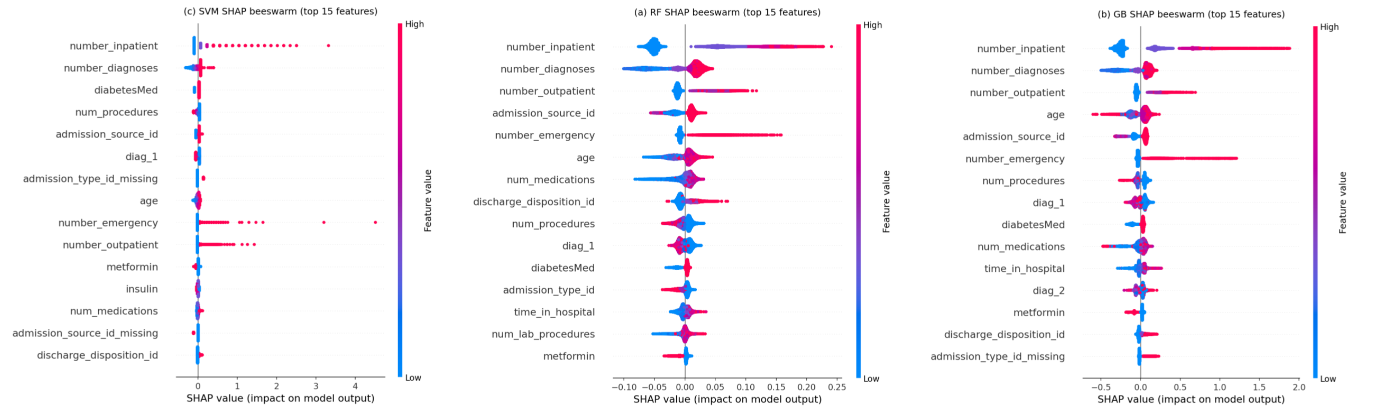

Figure 13: SHAP beeswarm plots for the top 15 features per model, computed on the 20,000-sample stratified subset. Each point represents one sample; horizontal position shows the SHAP value (impact on model output) and colour shows feature value (red = high, blue = low). **(a)** RF: `TreeExplainer`. **(b)** GB: `TreeExplainer`. **(c)** SVM: `LinearExplainer`. Features are ranked by mean |SHAP| value.

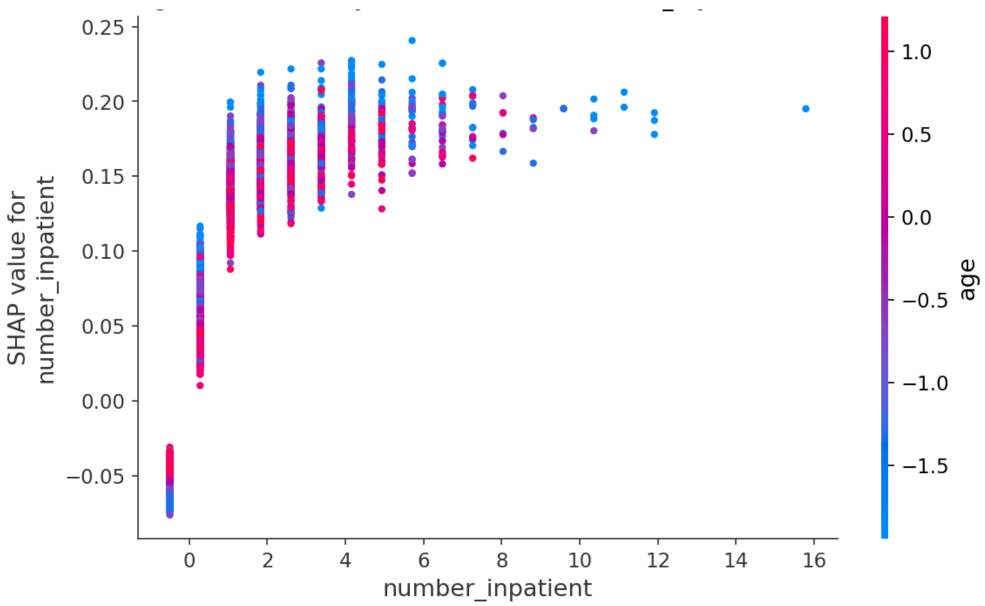

Figure 14: SHAP dependence plot for `number_inpatient`, the top-ranked RF feature by mean |SHAP|, computed on the 20,000-sample stratified subset using `TreeExplainer`. Each point represents one sample; the colour shows the value of the interaction feature selected automatically by SHAP. Horizontal position shows the raw feature value; vertical position shows the corresponding SHAP value.

4.3 Clinical Interpretation

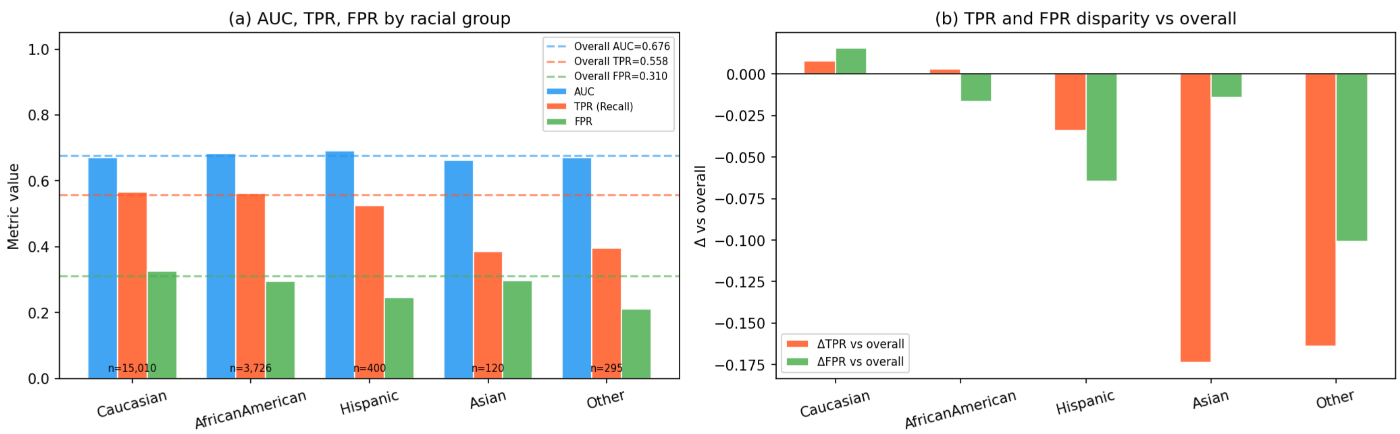

Figure 15: Fairness audit for the best RF model, pooled across all 5 outer folds at $\tau = 0.5$. Racial group labels are taken directly from the dataset. **(a)** AUC, TPR, and FPR per group; dashed horizontal lines show overall pooled values. Sample sizes are annotated at the base of each group. **(b)** ΔTPR and ΔFPR relative to the overall pooled values, assessing equalised-odds parity (threshold: $|\Delta| < 0.05$).

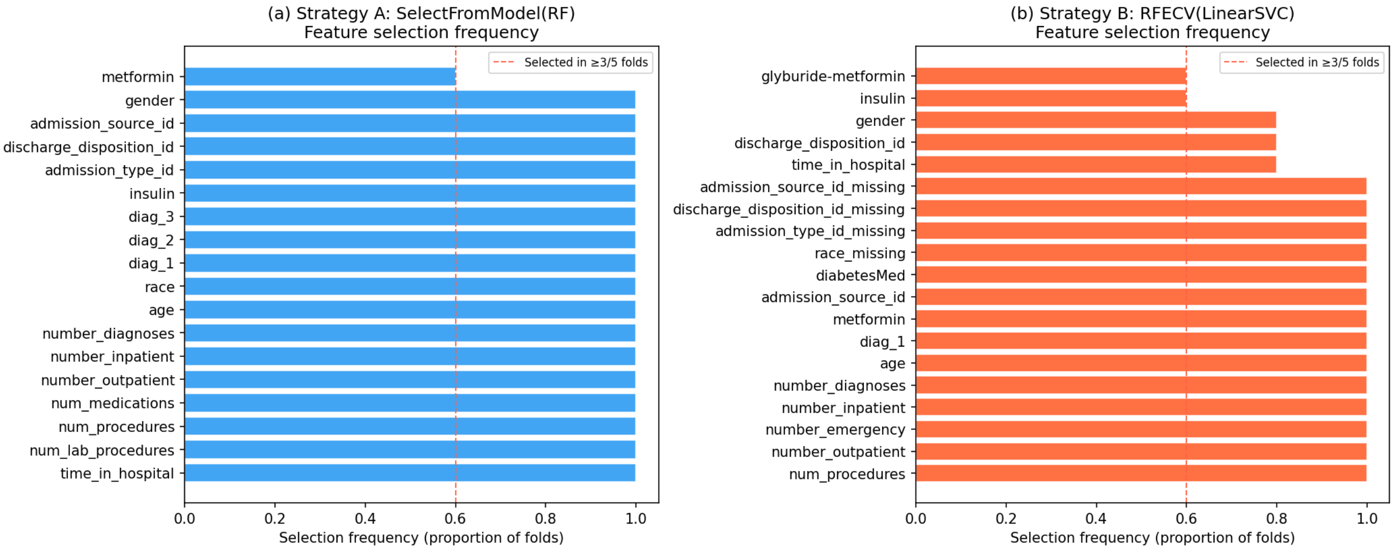

Figure 16: Feature selection frequency across 5 outer folds for features selected in at least 3 of 5 folds. The dashed vertical line marks the $\geq 3/5$ threshold. (a) Strategy A: `SelectFromModel(RF)`, 18 features shown. (b) Strategy B: `RFECV(LinearSVC)`, 19 features shown. Features absent from the plot were selected in fewer than 3 folds.